Pham Anh Cuong

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PROFILE

Machine learning researcher studying **efficient and trustworthy deep learning**, particularly how changes in representations affect knowledge retention and reliability during adaptation. My long-term goal is to learn latent representations that can be generalized across different applications.
Prior experience includes stochastic optimization, optimization-based continual learning, parameter-efficient fine-tuning and federated fine-tuning. Current research investigates uncertainty quantification of LoRA-based LLM fine-tuning for finance.

EDUCATION

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI), UAE — Aug 2024 - May 2026
Master of Science in Machine Learning

- **GPA:** 3.72/4.0 (∼94/100, with all courses grades above 90/100). [Transcript, 36 credits]
- **Thesis:** "Learning in the Null Space: Small Singular Values for Continual Learning" [PDF].
- Supervisors: Prof. Samuel Horváth, co-supervised by Prof. Praneeth Vepakomma.
- Selected courses: Mathematics for AI, Advanced ML, Probabilistic & Statistical Inference, Optimization.

VNU University of Engineering and Technology (VNU-UET), Vietnam — Aug 2019 - Jun 2023
Bachelor in Information Technology (Honors Program)

- **GPA:** 3.71/4.0 (8.78/10) - Graduated with High Distinction, top 5% CS Department. [Transcript, 146 credits]
- **Thesis:** "Building a Vietnamese image captioning dataset and developing a visual-text transformation model for generating image captions" (Thesis Grade: 9.5/10).
- Supervisors: Prof. Quang-Thuy Ha, Dr. Van-Quang Nguyen, MS. Thi-Hong Vuong.

RESEARCH EXPERIENCE

MBZUAI, Abu Dhabi, UAE
Research Engineer (Supervisor: Prof. Praneeth Vepakomma) — Jun 2026 - present

- **Project:** Uncertainty quantification of LoRA-based fine-tuning for finance (funded by ADIA Lab)
  Developing an uncertainty-aware LLM fine-tuning method that uses probabilistic view and statistical learning tools, also deals with multimodalities (text, time-series, tabular), with planned evaluation of predictive performance, calibration and failure detection before applying for financial tasks (first-author work, in progress).

Graduate Research Assistant (Supervisor: Prof. Samuel Horváth) — Oct 2024 - May 2026

- **Project:** Continual learning using low-rank updates (co-supervised by Prof. Praneeth Vepakomma)
  Developed an orthogonality-based LoRA-style update method that uses small singular value subspaces to reduce catastrophic forgetting while maintaining competitive model accuracy, resulting in a first-author oral paper [C.2] and currently expanding to continual fine-tuning (first-author work, in progress).
- **Project:** Parameter-free stochastic optimization (co-supervised by Prof. Martin Takáč)
  Co-developed an adaptive Polyak-style stepsize method based on twin iterates that removes the need for prior knowledge of the optimal objective value and contributed to the experimental implementation, leading to preprint [P.3].
- **Project:** Federated optimization and fine-tuning (co-supervised by Prof. Praneeth Vepakomma)
  Reproduced and analyzed Fed-SB and LoRA-SB under continual learning and multi-task learning setups.
- **Project:** Small language models for robotic path planning (extended from MBZUAI coursework project)
  Utilized SLMs with the guidance from LLMs to enable efficient and adaptive real-time inference on edge devices for autonomous robotics path-planning tasks, leading to preprint [P.2].

DSKT Laboratory at VNU-UET, Hanoi, Vietnam — Sep 2021 - Sep 2023
Student Research Intern

- Built a Vietnamese image-captioning dataset and evaluated it with a lightweight Transformer-based captioning model using both grid and region features, leading to preprint [P.1].
- Co-developed a memory-based continual relation extraction approach using prototype-augmented memory, resulting in the paper [C.1].

PUBLICATIONS

Conference Papers

[**C.2**] **Cuong Anh Pham**, Praneeth Vepakomma, Samuel Horváth: "Learning in the Null Space: Small Singular Values for Continual Learning" (CPAL 2026 - Oral Presentation)

[**C.1**] Quynh-Trang Pham Thi, **Anh-Cuong Pham,** Ngoc-Huyen Ngo and Duc-Trong Le: "Memory-Based Method using Prototype Augmentation for Continual Relation Extraction" (IEEE RIVF '22)

Preprints

[**P.3**] Farshed Abdukhakimov, **Cuong Anh Pham**, Samuel Horváth, Martin Takáč, Slavomir Hanzely: "Polyak Stepsize: Estimating Optimal Functional Values Without Parameters or Prior Knowledge" (arXiv:2508.17288)

[**P.2**] Quang P.M. Pham, Khoi T. N. Nguyen, Nhi H. Doan, **Cuong A. Pham**, Qinbo Sun, Weimin Qi, Kentaro Inui, Dezhen Song: "SmallPlan: Leverage Small Language Models for Sequential Path Planning with Simulation-Powered, LLM-Guided Distillation" (arXiv:2505.00831)

[**P.1**] **Anh-Cuong Pham**, Van-Quang Nguyen, Thi-Hong Vuong, Quang-Thuy Ha: "KTVIC: A Vietnamese Image Captioning Dataset on the Life Domain" (arXiv:2401.08100)

INDUSTRY EXPERIENCE

Viettel Networks, Hanoi, Vietnam — Oct 2023 - Jul 2024

Data Scientist

- Developed computer-vision pipelines for telecommunication maintenance tasks, using segmentation, classification, and image-processing methods; resulted in an internal tool to support network maintenance.
- Built predictive and data-processing workflows over datasets ranging from 10M to 1B records to optimize network-resource allocation.

GHTK Joint Stock Company, Hanoi, Vietnam

R&D Engineer — Aug 2022 - Aug 2023

- Co-developed and deployed an internal email spam classifier, achieving 90% F1-score on a corpus of ∼50,000 emails.

R&D Engineer Intern — May 2022 - Aug 2022

- Developed a Vietnamese document classification prototype using doc2vec during the internship program.

TALKS & PRESENTATIONS

"**Learning in the Null Space: Small Singular Values for Continual Learning**"

- Oral (9 papers selected) and Poster Presentation at CPAL (Tübingen, Germany) — 2026
- At RelationalML Group at CISPA, hosted by Prof. Rebekka Burkholz (online) — 2026

HONORS & AWARDS

Full scholarship for 2 years of the Master of Science program at MBZUAI — 2024
3 × Third Prizes (5th → 6th → 5th place) in Linear Algebra, VNU-UET Mathematical Olympiad — 2021, 2022, 2023
Yamada Scholarship (6 students/year at VNU-UET) — 2022
Academic Excellent Scholarship at VNU-UET (top 5 selected students) — 2021
Selected to represent Nghe An province (rank 9/10 selected students) in Vietnam Mathematical Olympiad — 2019

OTHER ATTENDED CONFERENCES, SCHOOLS & SEMINARS

Efficient Machine Learning Seminar at MBZUAI (Abu Dhabi, UAE) — 2025-present
(MLWS '26) MBZUAI Machine Learning Winter School on Representation Learning & GenAI (Abu Dhabi, UAE) — 2026
(ICOMP '25) The International Conference on Computational Optimization (Abu Dhabi, UAE) — 2025
(ASCOMP '25) The AI Autumn School of Computational Optimization (Abu Dhabi, UAE) — 2025

Language: English (IELTS 7.0), Vietnamese (native).

REFERENCES

Dr. Samuel Horvath, Assistant Professor at MBZUAI (UAE): (samuel.horvath@mbzuai.ac.ae)
Dr. Praneeth Vepakomma, Assistant Professor at MBZUAI & Visiting Professor at MIT: (vepakom@mit.edu)